

Agency MBS Weekly

HARP 1.5

- **Maintain modest overweight:** We maintain a modest overweight recommendation on the basis. 30y FNs have outperformed Treasury curve hedges by 5-7 ticks over the past week. A decline in global capital markets volatility has facilitated MBS spread tightening back to 2015 average levels. The change in tune at the Fed and the BOJ this week suggests further upside momentum to risk assets, which should also benefit MBS. With 10y JGB yields declining to all-time lows, foreign investor demand for MBS is likely to see an uptick, and broader market stability should also bode well for bank demand. We recognize, however, the fickle nature of the global market relief rally and recommend that investors pare down the overweight if MBS outperform further sharply, or 10y Treasury yields sink lower towards the 1.80-1.85% convexity zone. We maintain a neutral recommendation across the 30y stack and on G2/FNs.
- **Refi index average loan size hits multi-year high:** The conventional refi index increased another 12% this week, following a 21% increase last week. The strong rebound is partially caused by the non-conforming jumbo loans as the refi index increase is also accompanied by a multi-year high in refi loan size, a typical result of increasing jumbo refi share into a rally. The Freddie PMMS came in at 3.79% this week, 20bp away from the lowest level we saw in early 2015. A further rally will boost speeds particularly on loans originated in late 2014 and 1H 2015 as this will be the lowest rates these borrowers have seen since they are 6 month seasoned and servicers may now solicit these borrowers aggressively.
- **January issuance partially boosted by TRID:** January gross issuance currently stands at \$88bn, \$1bn lower than December and slightly higher than our projection of \$82bn. The large January gross issuance could lead to close to \$20bn net issuance based on our projection of a 20% decline in January speeds. We attribute part of the high purchase issuance to TRID, which resulted in more loan closing in December and issuance in January.
- **HARP 1.5:** We attempt to read the tea leaves on the streamline refinance program that FHFA Director Mel Watt hinted at last year and included in the 2016 GSE scorecard. The experience with various versions of HARP reveals the features that are likely to be on the FHFA wish list for the new streamline refi program. Some of them, however, could face hurdles such as GSE charter restrictions, PMI companies and risk-sharing transaction investors. We discuss how the GSEs can surmount most of these hurdles but may nevertheless aim for a slightly less aggressive program than the crisis-born HARP 2.0. In terms of borrower eligibility, a carve-out of HARPed borrowers is possible but beyond that it is likely that all vintages may be allowed to avail of the new program.

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Market Overview

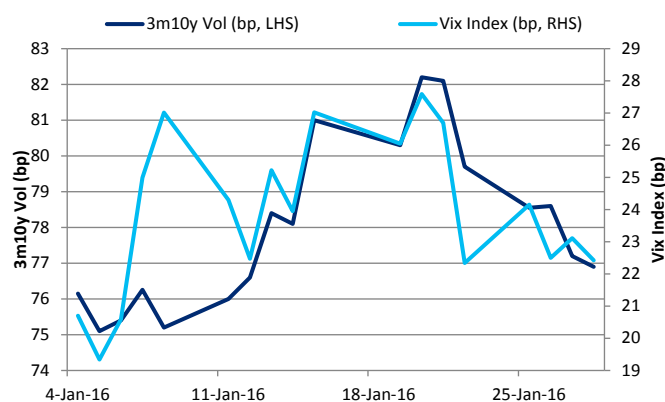
Figure 1. 30y FNs outperformed Tsy curve hedge by 5-7 ticks over the past week

	Performance WoW (tick)	OAS Chg (bp)	ZVOAS Chg (bp)
FNCL 3.0s	6.0	-2	-2
FNCL 3.5s	6.9	-3	-4
FNCL 4.0s	6.5	-3	-6
FNCL 4.5s	5.5	-3	-6

Source: Citi Research, Yieldbook

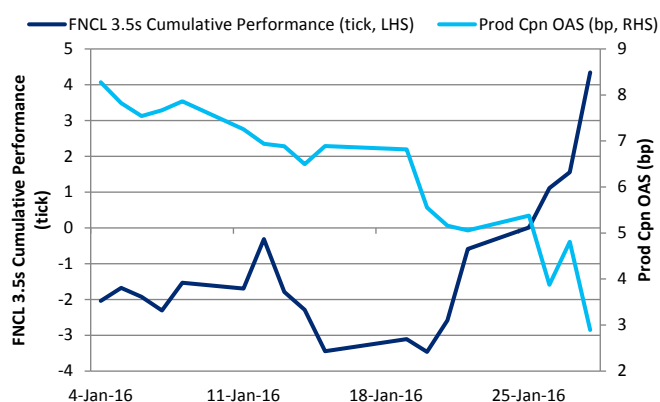
We maintain a modest overweight recommendation on the basis. 30y FNs have outperformed Treasury curve hedges by 5-7 ticks over the past week (Figure 1). A decline in global capital markets volatility (Figure 2) has facilitated MBS spread tightening back to 2015 average levels (Figure 3). The change in tune at the Fed and the BOJ this week suggests further upside momentum to risk assets, which should also benefit MBS. With 10y JGB yields declining to all-time lows, foreign investor demand for MBS is likely to see an uptick, and broader market stability should also bode well for bank demand. We recognize, however, the fickle nature of the global market relief rally and recommend that investors pare down the overweight if MBS outperform further sharply, or 10y Treasury yields sink lower towards the 1.80-1.85% convexity zone.

Figure 2. Asset volatility has declined recently...



Source: Citi Research, Bloomberg

Figure 3. ...helping MBS outperformance and spread tightening

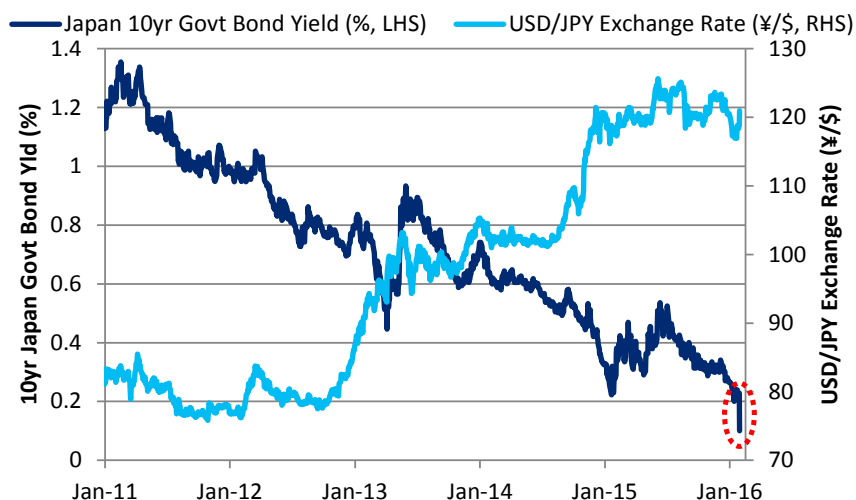


Source: Citi Research, Yieldbook, Bloomberg

Bad news is good news...

The recent bout of volatility in the global financial markets got a nod from central bankers this week. The BOJ introduced negative rates on new bank reserves earlier today, which should push Japanese investors towards dollar assets in search of yield as 10y JGB yields reached 10bp, their lowest level ever (Figure 4). The yen also depreciated against the USD sharply, suggesting the currency outlook will also favor investments in dollar markets. We should learn more about allocations from private investors as new year investment plans are released in a couple of months.

Figure 4. 10yr JGB yields reached 10bp, the lowest level ever

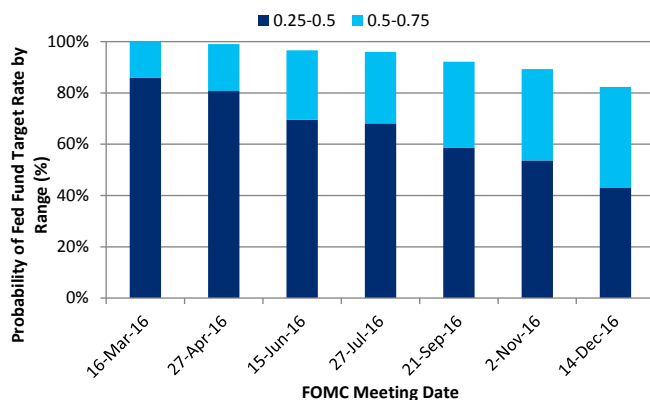


Source: Citi Research, Bloomberg

The January FOMC statement also leaned very dovish, although it stopped short of ruling out a rate move in March as per Citi economists ([US Economic View](#)). Fed funds futures market is now pricing in only a 14% chance of a rate hike in March, and the likelihood of Fed funds rate staying at current levels or being hiked just once through year end has increased to over 80% (Figure 5).

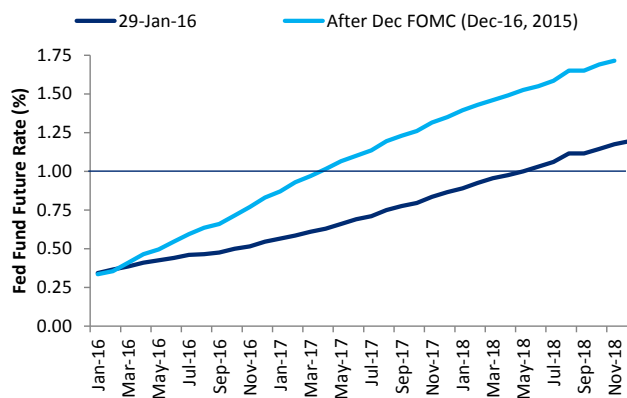
Looking further out the horizon the market pricing for Fed funds futures to reach 1% — a level we have used as a rough threshold for reinvestment tapering to begin — has recently been pushed into 2018 for the first time (Figure 6). This metric has seen some large shifts over the past few months, varying from early-2017 in December to Spring-2018 now. If the macro environment improves rapidly, we wouldn't be surprised if this timeline is pulled forward sharply once again. For the time being, however, reinvestment horizon has extended materially and should support MBS performance.

Figure 5. Fed funds futures market is pricing in a 80% probability of rates staying at current level or hiked just once through year end



Source: Citi Research, Bloomberg

Figure 6. Current market expectation for Fed Fund Futures to reach 1% has recently extended into 2018 for the first time

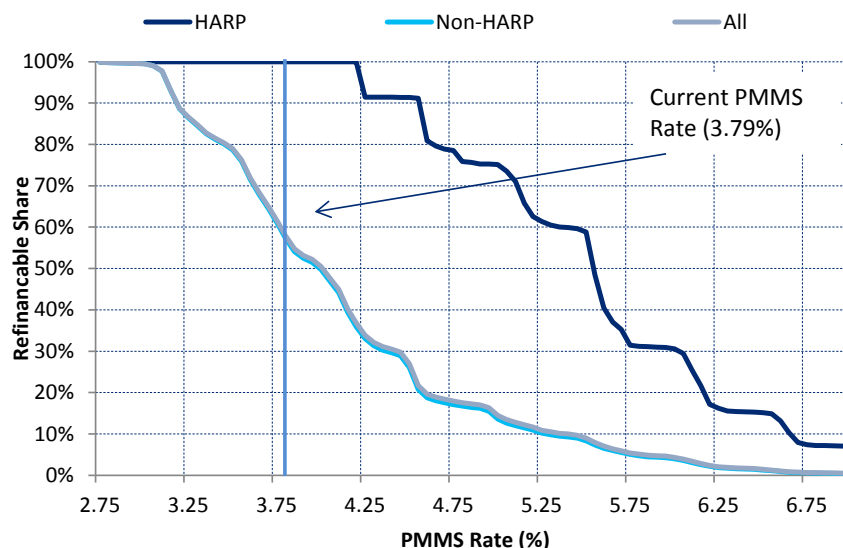


Source: Citi Research, Bloomberg

...but only for so long

The key risk emanating from the resetting of expectations about central bank actions is a sustained rally in long-term Treasury yields. 10y Treasury is trading at 1.92% yield at the time of writing, 7bp lower on the day. Freddie Mac survey reports primary mortgage rates at 3.79% as of Tue/Wed, when 10y was at 2%. A continued rally towards 1.8% on the 10y could take mortgage rates in the 3.5-3.625% range, which we have identified as a level for significant pickup in prepaids (Figure 7).

Figure 7. If 10y treasury further rallies towards 1.8% and mortgage rates is taken to 3.5-3.625%, prepayment will pick up significantly



Source: Citi Research, Freddie Mac, Bloomberg, 1010Data

FHA multifamily MIP cut highlights risk to rich valuations

Figure 8. The FHA announced a multifamily MIP reduction by category on January 28th:

	Upfront MIP Reduction	Annual MIP Reduction
Broadly Affordable	20-75bp	20-25bp
Affordable mixed-income	10-65bp	10-35bp
Energy-efficient	20-75bp	20-45bp
Market-rate	-	-

Source: Citi Research, HUD

The FHA announced a multifamily MIP reduction on January 28th that aims to encourage investment in affordable housing and energy-efficient apartments. The annual MIP cut ranged from 10-45bp and upfront MIP cut from 10-75bp for the various categories (Figure 8). Our view on the FHA MIP cut has been that it will result largely in a substitution of GSE high-LTV lending rather than new credit creation, but the recent improvement in the insurance fund reserve level provides the FHA leadership some ammunition to lower MIPs should they want to do so for political expediency.

The latest announcement of multifamily MIP reduction is too narrow in scope to draw any firm conclusion on the implications for a single-family MIP cut. The reductions this week refer to specific housing sectors that the administration has been trying to promote and that also found a place in the 2016 scorecard for the GSEs. On the margin though, this action raises the probability of a MIP cut on the single-family side and brings forth a risk that we have been highlighting for GNs in recent weeks given very rich valuations in that sector as we discuss below.

Summary and Valuations

We present a summary of our views across the agency MBS sectors in Figure 9.

Figure 9. Summary of views across agency MBS sectors

	Summary view	Reference Report
Basis	We maintain a modest overweight on the basis	Current report
Trade Recommendations	Long GD/FN 3.5s, Overweight HLB 4.0s	2016 Agency MBS Outlook - When the Tide Goes Out
Supply	Net supply of \$ \$125bn and gross supply of \$1.15 trillion in 2016. Gross issuance to be between \$80-90bn/month.	Agency MBS Weekly - 2016 Supply Outlook , Agency MBS Weekly
Demand	Banks to start buying while money managers could add to overweights. GSEs and REITs will reduce holdings gradually while overseas adds.	2016 Agency MBS Outlook - When the Tide Goes Out , Agency MBS Weekly
Speed Projections	30-yr FNCLs for Jan: -20% MoM; Feb: +3% MoM	Agency MBS Weekly, Jan 15, 2016
Coupon Stack	Up-in-coupon looks slightly cheaper, 3.5s face significant risks in a rally.	Current report
G2/FN	Neutral on G2/FN swaps	Agency MBS Weekly
Specified Pools	Loan balance 3.0s and 4.0s look attractive, we recommend long HLB 4.0s	US Agency MBS Focus - Overweight HLB 4.0s
15s/30s	Scale out of 15s/30s	2016 Agency MBS Outlook - When the Tide Goes Out
Hybrid ARMs	Quicken and tier 1 issued 5/1s and 7/1s look attractive to 15s for buy and hold investors	Agency MBS Weekly

Source: Citi Research

Further richening down in coupon, neutral on G2/FN stack

Figure 10 and Figure 11 show updated valuations on the 30y FNCL and G2SF coupon stack. As we had highlighted last week, FN 3s and 3.5s appear slightly rich in comparison to FN 4s, trading about 3bp tighter OAS. Over the past week, outperformance in FN 3s-4s against Treasury curve hedges was quite comparable at 6-7 ticks as it was led by a decline in global asset volatility. We continue to see significant downside risk to relative performance in 3.5s if rates rally given the coupon's high negative convexity and tight valuations. FN 3s offer a lower risk profile but their performance should be highly directional to rates. We recommend a neutral allocation across the stack in view of our expectation of continued macro-driven spread tightening.

Figure 10. Valuations on the 30-yr FNCL coupon stack

Coupon	WALA/WA C/ALS	Yield (%)	HAC (ticks)	zv- spread (bps)	OAS (bps)	Duration (yrs)	Convexity	1yr CPR (%)
FN 3.0s	8/3.76/322	2.69	3.1	66	0	5.2	-3.9	8
FN 3.5s	6/4.12/310	2.56	4.2	77	0	3.8	-4.5	13
FN 4.0s	17/4.56/281	2.40	4.7	76	3	2.5	-3.8	23
FN 4.5s	20/5.07/262	2.30	4.1	83	17	2.2	-2.8	25

Source: Citi Research, Yieldbook, as of Jan 28, 2016 Close

Figure 11. Valuations on the 30-yr G2SF coupon stack

Coupon	WALA/WA C/ALS	Yield (%)	HAC (ticks)	zv- spread (bps)	OAS (bps)	Duration (yrs)	Convexity	1yr CPR (%)
G2 3.0s	8/3.46/273	2.40	3.2	56	-18	4.6	-4.0	10
G2 3.5s	11/3.87/259	2.08	4.8	55	-21	3.1	-3.9	20
G2 4.0s	16/4.34/210	1.84	3.5	46	-15	2.1	-2.6	29
G2 4.5s	27/4.81/172	1.99	0.5	62	11	2.0	-2.1	31

Source: Citi Research, Yieldbook, as of Jan 28, 2016 Close

Figure 12 shows the spread and carry differential for G2/FN swaps. At 20bp tighter OAS than FNs, G2s offer little fundamental value amidst significant risks as the FHA multifamily MIP cut reminded the markets this week. We maintain a neutral allocation to G2s versus conventionals in view of positive market technicals for now (low supply, potential bank demand). If current valuations persist into the seasonal pickup in supply in the Spring, we will look to set up short position in the sector.

Figure 12. Spread and carry differential along with recommended positioning on G2/FN swaps

G2/FN	Yield	z-spread	OAS	HR	HAC	Recommended Positioning
3.0s	-0.30	-10	-19	90%	0.1	Neutral
3.5s	-0.48	-21	-21	82%	1.2	Neutral
4.0s	-0.56	-29	-18	86%	-0.6	Neutral

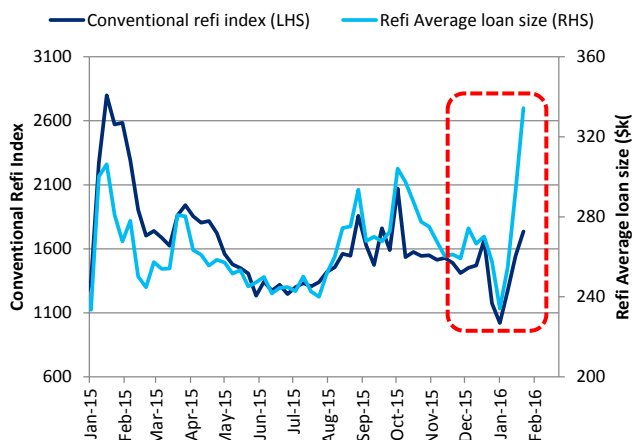
Source: Citi Research, Yieldbook, as of Jan 28, 2016 Close

Refi index average loan size hits multi-year high

The conventional refi index increased another 12% this week, following a 21% increase last week. The strong rebound is partially caused by non-conforming jumbo loans as the refi index increase is also accompanied by a multi-year high in refi loan size, a typical result of increasing jumbo refi share into a rally (Figure 13). We have noticed an increasing gap between jumbo rates and conforming rates recently (Figure 14). Based on Wells Fargo's online quote, refi rates for jumbo loans used to be on par or only marginally lower than conforming loans but the gap has increased to negative 20bp recently, further worsening the callability of jumbo loans relative to conforming loans.

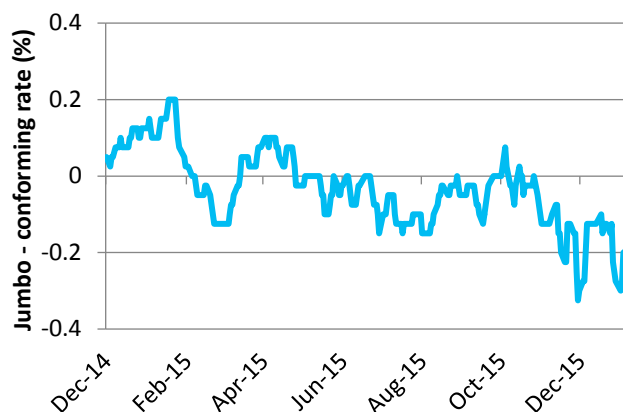
Although the current refi index is only slightly higher than levels observed in October-November last year, with 10yr currently at the lowest level since May 2015, refi risk has increased materially. The Freddie PMMS came in at 3.79% this week, 20bp away from the lowest level we saw in early 2015. A further rally will boost speeds particularly on loans originated in late 2014 and 1H 2015 as this will be the lowest rates these borrowers have seen since they are 6 month seasoned and servicers may now solicit these borrowers aggressively.

Figure 13. The average refi loan size hit multi-year high while the index is much lower than early last year, a sign of high jumbo share



Source: Citi Research, Bloomberg

Figure 14. 30y Jumbo refi rate is now 20bp lower on average compared to conventional rate (5-day average)



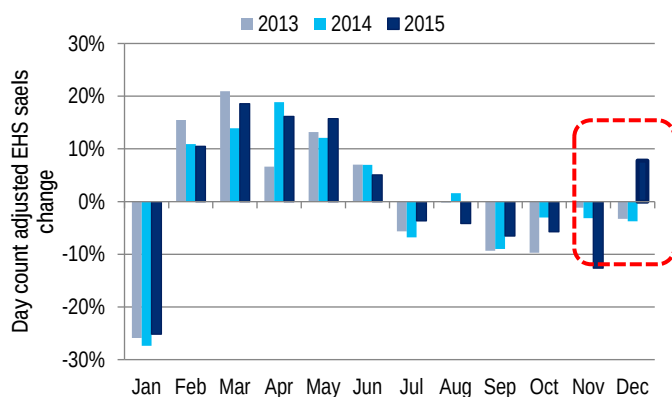
Source: Citi Research, Citi MQA, Wells Fargo

January issuance partially boosted by TRID

January gross issuance currently stands at \$88bn, \$1bn lower than December and slightly higher than our projection of \$82bn (see Jan. 8th [Weekly](#)). The large January gross issuance could lead to close to \$20bn net issuance based on our projection of a 20% decline in January speeds.

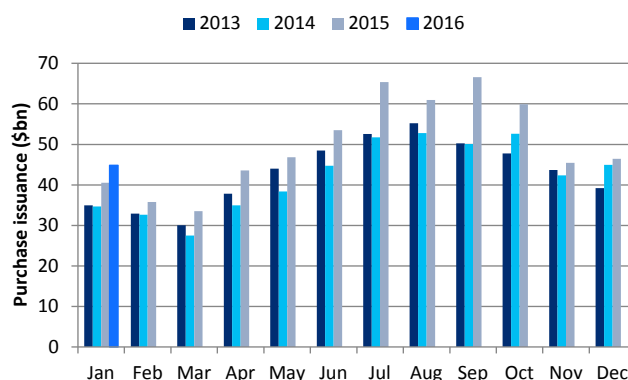
We attribute part of the increase to TRID which resulted in more loan closings in December and hence MBS issuance in January (Figure 15, Figure 16). Recall that TRID-lengthened loan closing time had led November EHS to drop and December EHS to increase much more than historical home sale seasonals would suggest. We expect February gross and net issuance to drop from here and Q1 overall supply to remain muted.

Figure 15. Day count adjusted EHS dropped in November and rebounded in December



Source: Citi Research, Bloomberg

Figure 16. Strong EHS in December boosted purchase issuance in January



Source: Citi Research, eMBS

HARP 1.5

We attempt to read the tea leaves on the streamline refinance program that FHFA Director Mel Watt hinted at last year and included in the 2016 GSE scorecard. The experience with various versions of HARP reveals the features that are likely to be on the FHFA wish list for the new streamline refi program. Some of them, however, could face hurdles such as GSE charter restrictions, PMI companies and risk-sharing transaction investors. We discuss how the GSEs can surmount most of these hurdles but may nevertheless aim for a slightly less aggressive program than the crisis-born HARP 2.0. In terms of borrower eligibility, a carve-out of HARPed borrowers is possible but beyond that it is likely that all vintages may be allowed to avail of the new program.

Reading between the lines of FHFA-speak

The 2016 scorecard for the GSEs requires them to “complete assessment and, as appropriate, begin development of a high loan-to-value ratio refinance program to ensure a January 2017 implementation”. This requirement comes in the context of HARP expiry at the end of this year. However, it is not clear what the new program will entail and which borrowers it will target. This has caused significant investor anxiety as in the worst case scenario it could allow re-HARPing, which will impact payups on MHA/CQ/CR paper. At the very least even **a prospective streamline refi program will worsen the convexity of mortgages**, particularly under a HPA downturn scenario.

FHFA Director Mel Watt’s speech in May 2015 provided the initial context for this initiative. He said:

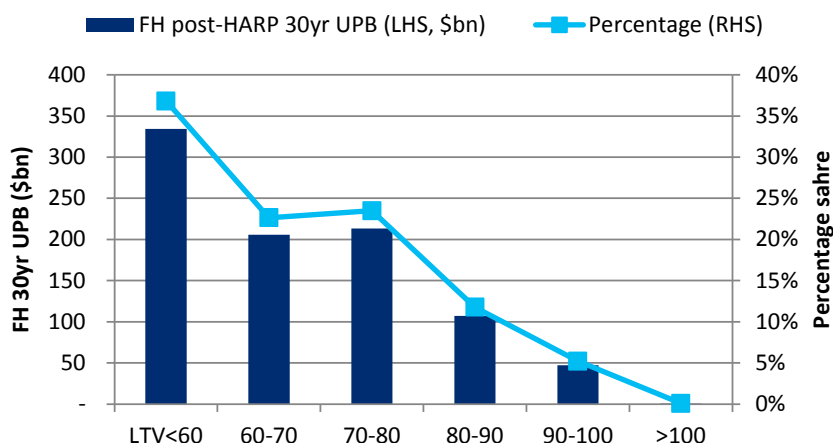
For HARP, we are also going to use this time to explore possible streamlined refinance solutions for future Enterprise loans. This evaluation will consider the lessons learned from the HARP program and how those might apply in a non-crisis environment.

With only that vague statement from seven months ago to guide the markets for now, we highlight a few possibilities:

- **Prospective or retroactive?:** Watt mentioned that the streamline refinance solution will be "for future enterprise loans", which implies that pre-Jan 2017 loans may not be eligible for this program. This is obviously the best case scenario for MBS investors as it does not impact current holdings. However, the 2016 scorecard does not repeat that language, suggesting that coverage of older vintages may still be on the table. Specifically, if the goal of the program is loss-mitigation and HARP replacement (the scorecard section in which the new program was listed as a goal), it is not obvious why any borrower would be excluded. A carve-out of HARPed borrowers is possible given FHFA's unwillingness to extend the HARP cutoff date in the past, but beyond that we think it is likely that all vintages may be allowed to avail of this program in the future.
- **Program features:** As Watt's statement stressed on considering the lessons learned from the HARP program, we think the new streamline refi will try to imitate the latest, most successful, version of HARP 2.0 to a large degree. In the next section we discuss the evolution from the pre-HARP streamline refinance program to HARP 1.0 to HARP 2.0 and then further improvements to the program from there on. We evaluate which features may be the easiest and most desirable for the GSEs to retain and the hurdles that they could face in retaining some of them.
- **How high LTV? :** Watt's original statement about applying the lessons learned from HARP "in a non-crisis environment" can be interpreted in two very different ways. On the one hand it could mean a streamline refinance for all borrowers, including low-LTV, as a regular course of business. In that case, we can expect a low-LTV program to follow the high-LTV program. This will broaden the impact of the streamline refi program significantly, as a vast majority of post-HARP borrowers are <80 LTV (Figure 17) and will be able to access this program.

On the other hand, the FHFA had waived the GSE charter requirement for mortgage insurance on >80 LTV loans citing 'unusual and exigent market circumstances' (a crisis). It is possible then that Watt was indicating a high-LTV streamline refinance program without MI grandfathering, but including other features from HARP. This would however leave the new program quite toothless for borrowers above 97 LTV currently as MI is not available. Also, in time of significant home price depreciation MI companies may not be willing to underwrite any >80 LTV borrowers, making this truly a solution for only the "non-crisis environment". We discuss the scenarios around this later in this report.

Figure 17. Only 15% of post-HARP Freddie 30yr loans are high LTV loans (excluding U6 and U9)



Source: Citi Research, 1010data

The above range of possibilities highlights the uncertainty about the outcome. The good news, however, is that the FHFA under Mel Watt has generally sought feedback from all stakeholders before implementing any key changes. In deciding on G-fee/LLPA changes, they sought public comment and have worked with originators behind the scenes in forging the new Rep & Warrant requirements too. In that vein, we expect the FHFA/GSEs to reach out to various stakeholders before the program is implemented and given the timeline of implementation, it is likely to happen within the next 2-4 months.

Lessons learned from HARP

Figure 18 summarizes the evolution of streamline refinance programs at the GSEs over the years. The biggest change that HARP 1.0 introduced to the pre-crisis streamline refi program was the **grandfathering of mortgage insurance coverage for borrowers up to 105 LTV initially (extended up to 125 LTV in July 2009)**. However, the program fell well short of expectations with just under 1mm borrowers taking advantage in the first two years compared to the 4-5 million eligible borrowers that the Treasury had estimated when introducing the program.

The introduction of **HARP 2.0** sought to address several key impediments to the success of the program. The five most notable changes and reasons for making them, as identified in a report by the FHFA OIG in Aug 2013, were:

1. Removal of the **125 LTV ceiling** to include more seriously underwater borrowers,
2. Extension of the program by **18 months** to encourage lenders to commit resources to the program,
3. Elimination of the manual property **appraisal** requirement to streamline the refi process and lower borrower costs,
4. Lower the **LLPA to a maximum of 75bp** for HARP borrowers, and
5. Revise borrower solicitation guidelines and **allow lenders to offer additional incentives** to borrowers.

Figure 18. Key differences in streamline refi, HARP 1.0 and HARP 2.0 programs

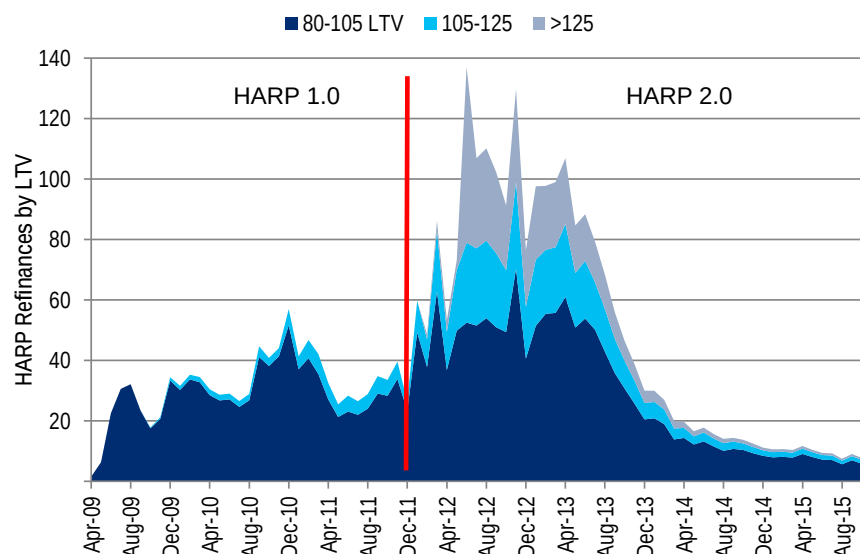
	HARP 2.0	HARP 1.0	Streamlined Refi
Period	Since Nov.2011	Apr.2009-Oct.2011	Pre-HARP 1.0
Appraisal	Mostly Automated	Manual/automated appraisal but manual appraisal required in most cases	Manual appraisal
Income & Asset documentation	Reduced requirement in Sep. 2012--same servicer are required to verbally verify a borrower's income; New lenders are required to have documentation	Need documentation In some cases need two year history of income	Stated Income/employment
Rep & warrant	Further relief of rep & warrant for original loan in Sep. 2012; Relief of new loan rep & warrant risk with 12 month clean pay following origination	Seller retains all the original rep & warrant regarding value condition and marketability; no additional relief on the new loan	Any previous rep & warrant relief on original loans does not apply to new loan
LTV restriction	Extended to above 125 LTV in Oct. 2011	LTV ceiling at 105 when announced in Mar. 2009 Raised ceiling from 105 to 125 in Jul. 2009	under 95-100
MI	MI transfers made easier	No new MI required but transfer of MI is cumbersome	Standard
LLPA	Lowered LLPA to 75bp max	Standard	Standard

Source: Citi Research, FHFA, Fannie Mae, Freddie Mac

Even as HARP 2.0 was extremely successful with a large increase in number of borrowers refinanced (Figure 19), further program changes were made over the course of 2012-13 based on lender feedback, which added to the program's success:

1. Rep & warrant relief after 12 months of clean pay history for the refinanced loan. Previously, lenders (same servicer as well as cross-servicer) were hesitant to take on the R&W risk for a new high-LTV loan as it was generally perceived that putback risk was the highest if a loan defaulted within the first two years of origination.
2. Lower income and asset documentation requirements
3. Alignment of same servicer and cross-servicer HARP refi requirements, and
4. Alignment of Fannie Mae and Freddie Mac guidelines
5. A further two-year extension of the program to December 2015, followed by another extension last year to December 2016.

Figure 19. Monthly HARP refinance volume by LTV (Thousands)



Source: Citi Research, 1010data

Baseline for the streamline refi program

Figure 20 classifies the major lessons learned in HARP execution into two categories – those that should be easy to include in the new streamline refinance program and others that may face hurdles. The changes on the left are largely at the FHFA/GSE discretion and contributed heavily to the success of HARP 2.0. Therefore, we expect to see most, if not all, of these features in the new streamline refi program. Rep & Warrant relief may be provided explicitly as under HARP, or could be addressed through the new R&W framework that the GSEs have been developing.

Figure 20. Features from HARP 2.0 that may be included in the new streamline refi program

Lessons learned in HARP Execution

Easy to include	May face hurdles
Long-term program availability	Lower LLPA
Elimination of manual property appraisal and low doc requirements	Mortgage Insurance grandfathering
Allow lender solicitation	No LTV ceiling
Rep & Warrant relief	
Alignment of same servicer and cross-servicer requirements	
Alignment of Fannie Mae and Freddie Mac guidelines	

Source: Citi Research

Willingness and ability to grandfather mortgage insurance

The features on the right hand side of Figure 20 above will be more nuanced to incorporate in the new program as they are not all solely under the FHFA/GSE discretion. For instance, MI grandfathering will require the cooperation of PMI companies. It is, however, in the PMI companies' interest to allow insured high-LTV borrowers to refinance into a lower payment, lowering the borrowers' chance of default, and therefore we think they are quite likely to agree.

Then there is the issue of the GSE charter requirement that >80 LTV loans should carry mortgage insurance. We think the FHFA has the power to override that

requirement given historical precedence. A February 20, 2009 letter from then FHFA Director James Lockhart to the EVP of Mortgage Insurance Companies of America laid down the foundation for that action¹. Fannie Mae's HARP FAQ document also references this letter in answering a question about charter override on MI coverage for >80 LTV loans, explaining how FHFA's interpretation of the Charter allows this for loans that the GSE already owns or guarantees.

The two key explanations advanced in that letter were the lack of refinancing opportunities for high LTV borrowers, and the lowering of credit risk for GSEs on loans they already guaranteed/owned by refinancing them into a lower rate. The below excerpt from that letter sums it up nicely:

In layman's term, the refinance initiative is akin to a loan modification for charter purposes as it affects loans for which an Enterprise already holds the credit risk. By creating an avenue for the borrower to reap the benefit of lower mortgage rates in the market, the credit risk of that mortgage to the Enterprise diminishes; thus, this is a loss-mitigation initiative in this very troubled time in housing finance. It has the added benefit of helping many households strengthen their own financial situation and enhance their commitment to their home and community.

Given this precedent, while the FHFA as conservator of the GSEs appears to have the power to override the MI requirement, it is not entirely clear that they will be willing to do this under the "non-crisis environment" that Watt mentioned in his speech. If not, this will render a high-LTV program quite toothless in a crisis environment and the "No LTV ceiling" feature may be explicitly excluded or rendered largely ineffective.

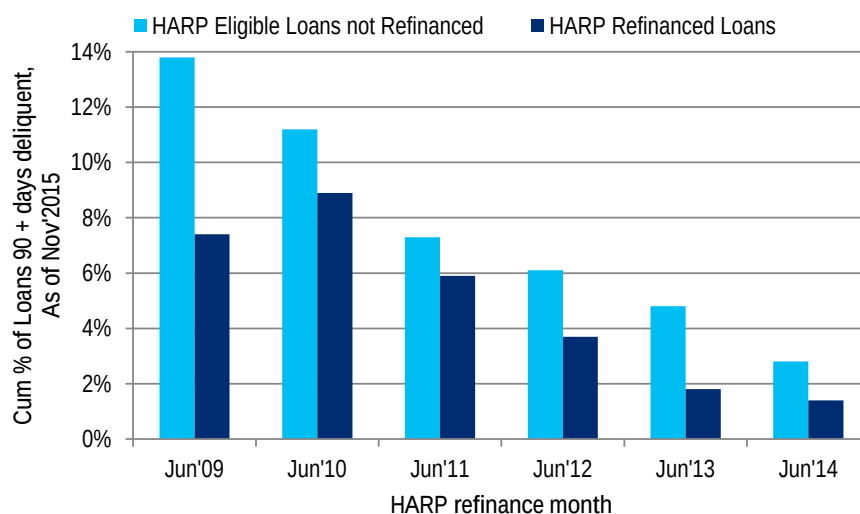
GSEs propose, CAS/STACR investors dispose?

The risk-sharing transactions that the GSEs have undertaken in recent years also create a complication for replicating the HARP 2.0 features. Firstly, the GSEs don't entirely own the credit risk for these loans so the argument above for MI grandfathering may not apply as directly to these loans. Secondly, and even more importantly, if the loans referenced in a CAS/STACR deal are refinanced, then the credit risk comes back to the GSEs under the current transaction structure. In a declining home price environment, this will effectively require the GSEs to take over the default risk when it is the highest. If so, the GSEs will likely require very high LLPAs to guarantee these loans, which will reduce the incentive significantly and could work against the success of the program.

A better option that the GSEs may explore will be to structure the risk-sharing transactions going forward with terms that require investors to retain the risk on a streamline refinanced high-LTV loan. This should be agreeable to lower-tranche investors as high-LTV streamline refinances will lower their credit losses in a negative HPA environment as HARP has established clearly in the recent years (Figure 21). Their situation is in effect similar to the PMI companies, which agreed to assign MI to new loans under HARP. For the higher-tranche investors similar benefits apply but the duration extension may lead to slightly more spread requirement.

¹ The letter is available at
<http://www.fhfa.gov/Media/PublicAffairs/PublicAffairsDocuments/HutchinsonGSERefi22009.pdf>.

Figure 21. Loans refinanced through HARP have shown lower cumulative 90+ ever delinquent rate than loans eligible but did not refinance through HARP



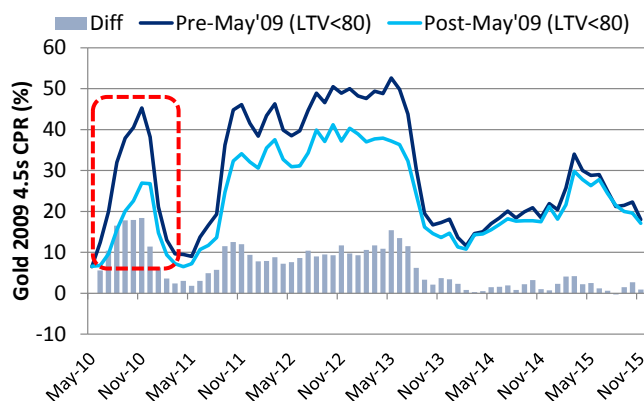
Source: Citi Research, FHFA

The (speed) devil is in the details

The significant contrast in speeds during HARP 1.0 versus HARP 2.0 makes it clear that the speed impact will depend on the features of the program. As discussed above, we think most of the procedural hurdles faced during HARP 1.0 are likely to be addressed in the streamline refinance program. However, if the GSEs require higher LLPA's to refinance these loans then it could lower borrower response significantly.

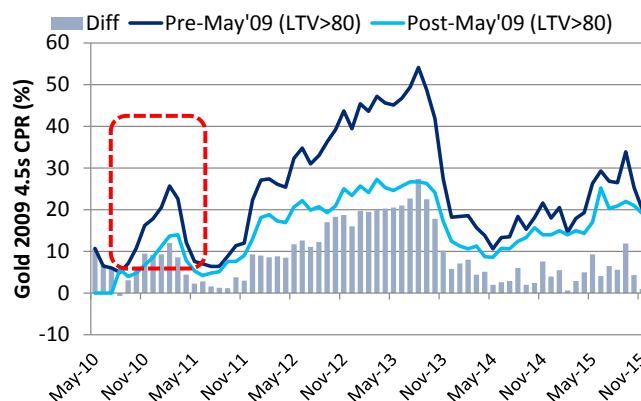
Figure 22 and Figure 23 show the speeds of Gold 2009 4.5s during HARP 1.0 versus HARP 2.0 for <80 LTV and >80 LTV borrowers, respectively. In HARP 1.0 high LTV borrowers faced high LLPA charge and lenders were more concerned with rep & warrant risk for high LTV borrowers. Therefore, speeds on high LTV loans were much slower compared to low LTV borrowers. As these additional hurdles were gradually resolved during HARP 2.0, high LTV speeds reached comparable levels as low LTV loans.

Figure 22. Low-LTV pre-May '09 borrower speeds jumped during HARP 1.0 itself...



Source: Citi Research, 1010data

Figure 23. ...but high LTV speeds were more muted due to frictions in HARP 1.0, and caught up to low-LTV in HARP 2.0



Source: Citi Research, 1010data

Rocket Fuel

Last week we had discussed how the advent of online mortgages such as Quicken's Rocket Mortgage could increase the efficiency of borrower's exercising their refi option as they make the process less cumbersome (see [Agency MBS Weekly - Escape Velocity](#)). Streamline refis, when implemented, will add another dimension to that process. While the front-end interaction between the borrower and lender is made easier with online mortgages, the back-end processing between the lender and GSEs is highly smoothed in streamline refis. If streamlined refis are extended to low-LTV borrowers, they could add even further to the risk of MBS convexity worsening in the future with faster speeds and shorter lags.

Appendix A-1

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